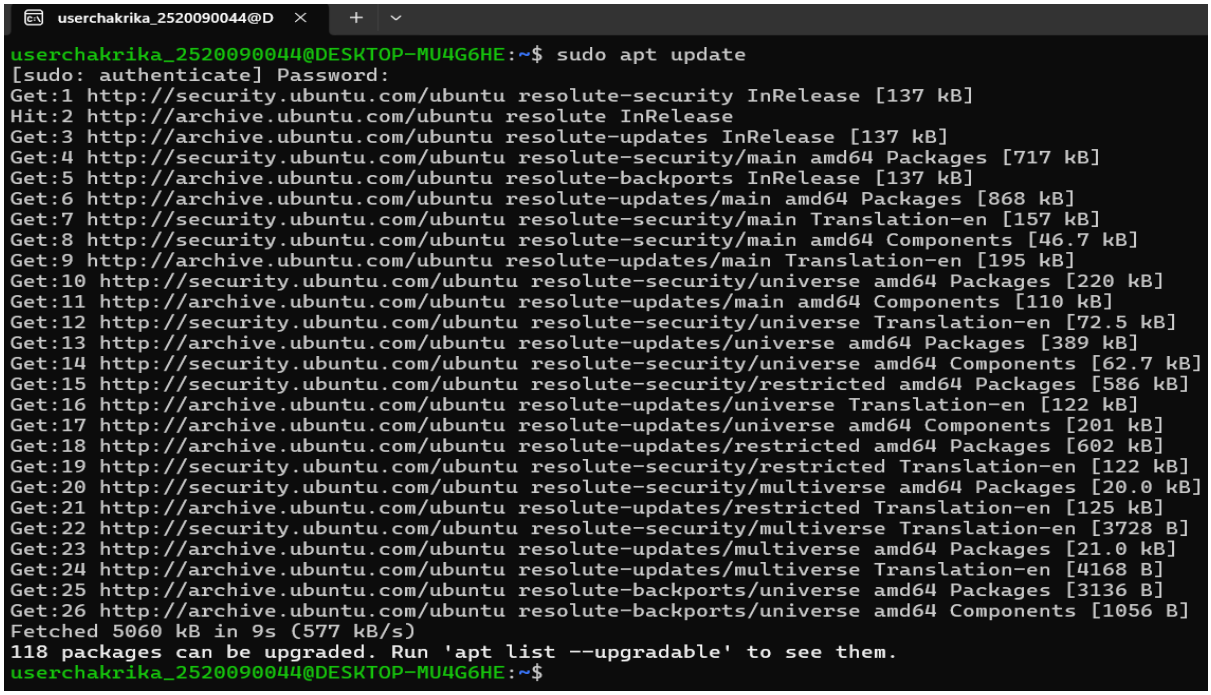
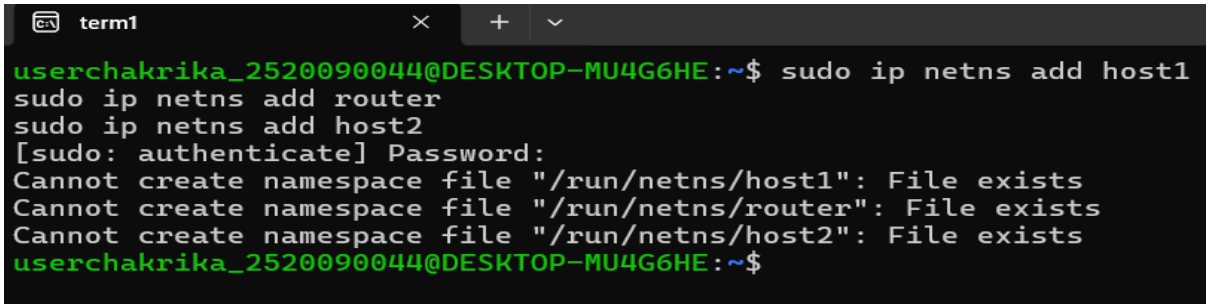
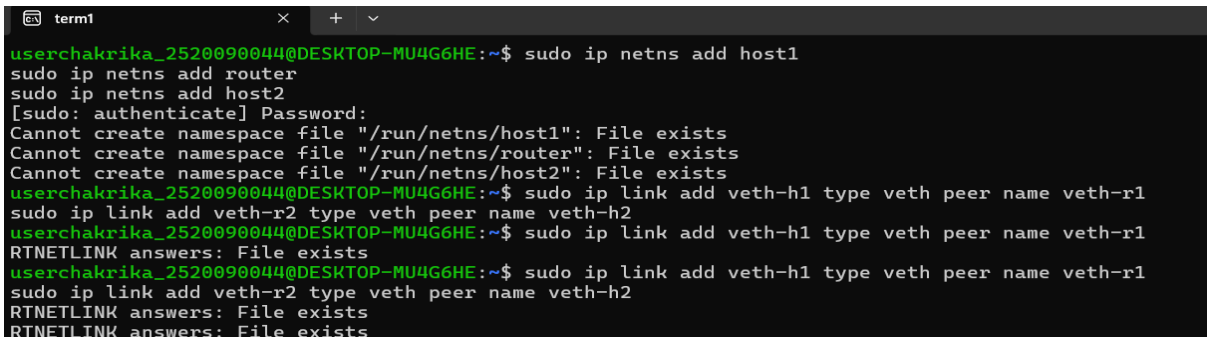
Skill-2(2520090044-Chakrika)

STAGE 1 — Install tools



STAGE 2 — Build the network





```
term1
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
[sudo: authenticate] Password:
Cannot create namespace file "/run/netns/host1": File exists
Cannot create namespace file "/run/netns/router": File exists
Cannot create namespace file "/run/netns/host2": File exists
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip link add veth-h1 type veth peer name veth-r1
RTNETLINK answers: File exists
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
RTNETLINK answers: File exists
RTNETLINK answers: File exists
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
userchakrika_2520090044@DESKTOP-MU4G6HE:~$
```

STAGE 3 — Configure IP addresses

```
term1
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
[sudo: authenticate] Password:
Cannot create namespace file "/run/netns/host1": File exists
Cannot create namespace file "/run/netns/router": File exists
Cannot create namespace file "/run/netns/host2": File exists
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip link add veth-h1 type veth peer name veth-r1
RTNETLINK answers: File exists
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
RTNETLINK answers: File exists
RTNETLINK answers: File exists
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
userchakrika_2520090044@DESKTOP-MU4G6HE:~$
```

STAGE 4 — Enable routing

```
sudo ip netns exec host2 ip link set lo up
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
userchakrika_2520090044@DESKTOP-MU4G6HE:~$
```

STAGE 5 — Test connectivity

```
net.ipv4.ip_forward = 1
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.075 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.044 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.043 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2028ms
rtt min/avg/max/mdev = 0.043/0.054/0.075/0.014 ms
```

STAGE 6 — Add delay and loss

```
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1%
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 13 limit 1000 delay 20ms loss 1% seed 17643234911818120117
userchakrika_2520090044@DESKTOP-MU4G6HE:~$
```

STAGE 8 — Start three captures simultaneously

Terminal 1 — capture at host1

```
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
```

Terminal 2 — capture at router

```
term2
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
```

Terminal 3 — capture at host2

```
term3
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ unbuntu
unbuntu: command not found
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
```

STAGE 9 — Send real message from host1 to host2

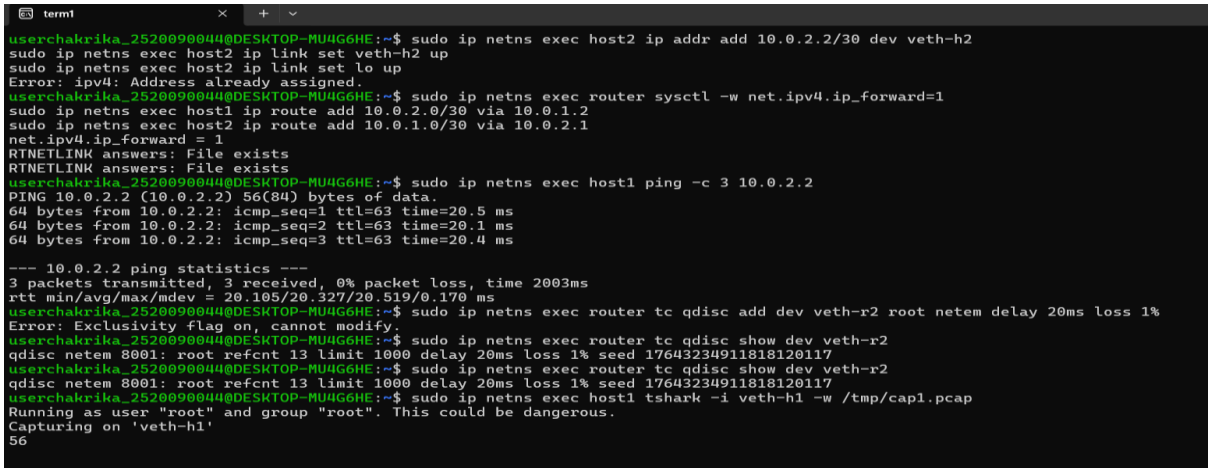
STAGE 10 — What you see in real time

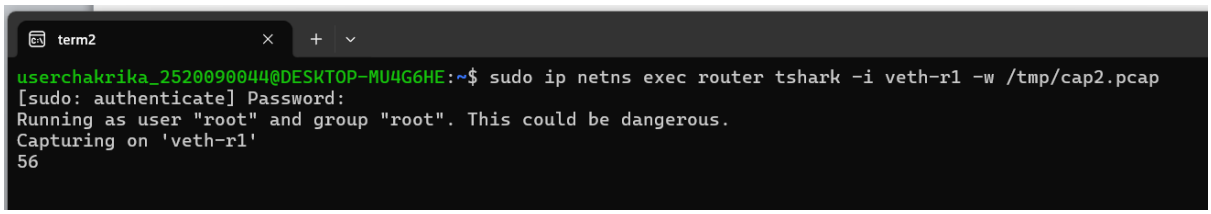
```
term4
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec host2 nc -l 1234
[sudo: authenticate] Password:

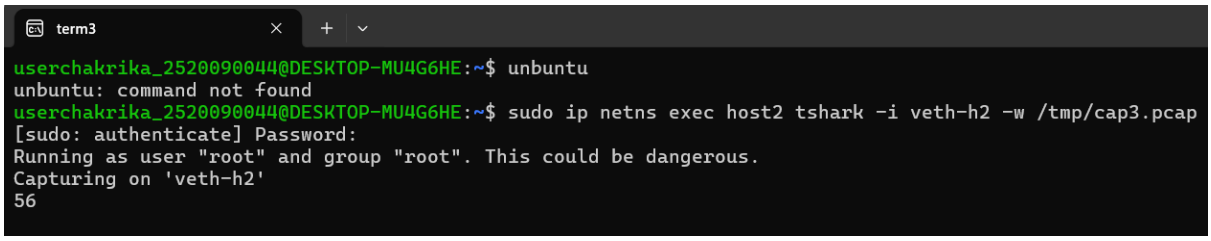
^Z
[1]+  Stopped                 sudo ip netns exec host2 nc -l 1234
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec host2 nc -l 1234
^Z
[2]+  Stopped                 sudo ip netns exec host2 nc -l 1234
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec host2 nc -l 1234
Hello from host1!
This is Alekhya
Computer Networks Lab
Three Point Capture Experiment
Message traveling through router
```

```
term5
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec host1 nc 10.0.2.2 1234
[sudo: authenticate] Password:
Hello from host1!
This is Alekhya
Computer Networks Lab
Three Point Capture Experiment
Message traveling through router
```

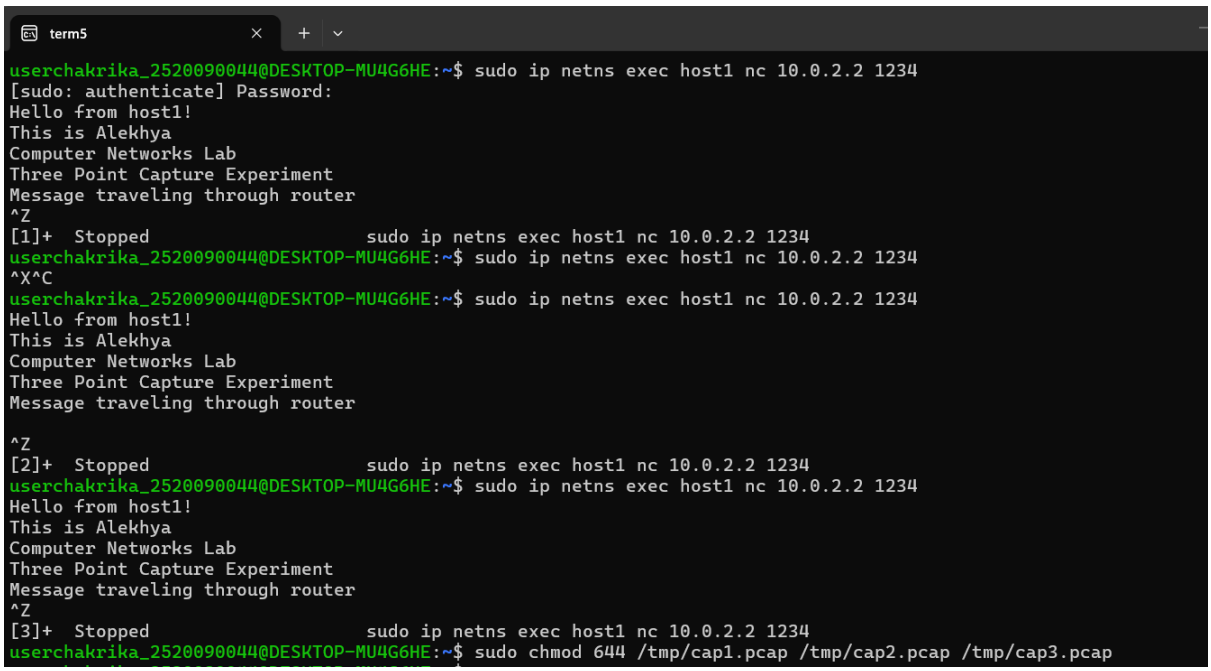
Stage 11: Terminal 1, 2, 3 (tshark) show packet counts increasing:







STAGE 12 — Fix permissions



STAGE 13 — Check TTL

```
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uni
q -c
tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
     18 63
     22 64
     18 63
     22 64
     22 63
     18 64
```

STAGE 14 — Read your actual message from the capture

```
Hello from host1!
This is Alekhya
Computer Networks Lab
Three Point Capture Experiment
Message traveling through router
```

Connect from host1:

```
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo apt install -y tshark iproute2 netcat-openbsd wireshark
[sudo: authenticate] Password:
tshark is already the newest version (4.6.4-1).
iproute2 is already the newest version (6.19.0-1ubuntu1.1).
netcat-openbsd is already the newest version (1.234-1).
wireshark is already the newest version (4.6.4-1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 64
userchakrika_2520090044@DESKTOP-MU4G6HE:~$
```

Create 3 nodes:

```
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
Cannot create namespace file "/run/netns/host1": File exists
Cannot create namespace file "/run/netns/router": File exists
Cannot create namespace file "/run/netns/host2": File exists
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns list
host2 (id: 2)
router (id: 1)
host1 (id: 0)
userchakrika_2520090044@DESKTOP-MU4G6HE:~$
```

Verify:

```
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns list
host2 (id: 2)
router (id: 1)
host1 (id: 0)
```

Create cables:

```
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
RTNETLINK answers: File exists
RTNETLINK answers: File exists
userchakrika_2520090044@DESKTOP-MU4G6HE:~$
```

Plug cables into nodes:

```
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
Error: An interface with the same name exists in the target netns.
Error: An interface with the same name exists in the target netns.
Error: An interface with the same name exists in the target netns.
Error: An interface with the same name exists in the target netns.
userchakrika_2520090044@DESKTOP-MU4G6HE:~$
```

Configure host1:

```
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
Error: ipv4: Address already assigned.
userchakrika_2520090044@DESKTOP-MU4G6HE:~$
```

Configure router:

```
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
Error: ipv4: Address already assigned.
Error: ipv4: Address already assigned.
userchakrika_2520090044@DESKTOP-MU4G6HE:~$
```

Configure host2:

```
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
Error: ipv4: Address already assigned.
userchakrika_2520090044@DESKTOP-MU4G6HE:~$
```

Enable routing:

```
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
RTNETLINK answers: File exists
RTNETLINK answers: File exists
userchakrika_2520090044@DESKTOP-MU4G6HE:~$
```

Test connectivity:

```
RTNETLINK answers: File exists
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=21.0 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=20.3 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=20.2 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 20.198/20.486/20.952/0.332 ms
userchakrika_2520090044@DESKTOP-MU4G6HE:~$
```

Add delay and loss Verify:

```
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 13 limit 1000 delay 20ms loss 1% seed 17643234911818120117
userchakrika_2520090044@DESKTOP-MU4G6HE:~$
```

TERMINAL 1 — Capture at host1

```
qdisc netem 8001: root refcnt 13 limit 1000 delay 20ms loss 1% seed 17643234911818120117
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
```

TERMINAL 2 — Capture at router

```
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
```

TERMINAL 3 — Capture at host2

```
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
```

TERMINAL 4 — Send the message

```
term4

userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec host2 nc -l 1234
[sudo: authenticate] Password:
```

```
term4

[1]+  Stopped                 sudo ip netns exec host2 nc -l 1234
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec host2 nc -l 1234
^Z
[2]+  Stopped                 sudo ip netns exec host2 nc -l 1234
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec host2 nc -l 1234
Hello from host1!
This is Alekhya
Computer Networks Lab
Three Point Capture Experiment
Message traveling through router
```

Check files are not empty:

```
term4

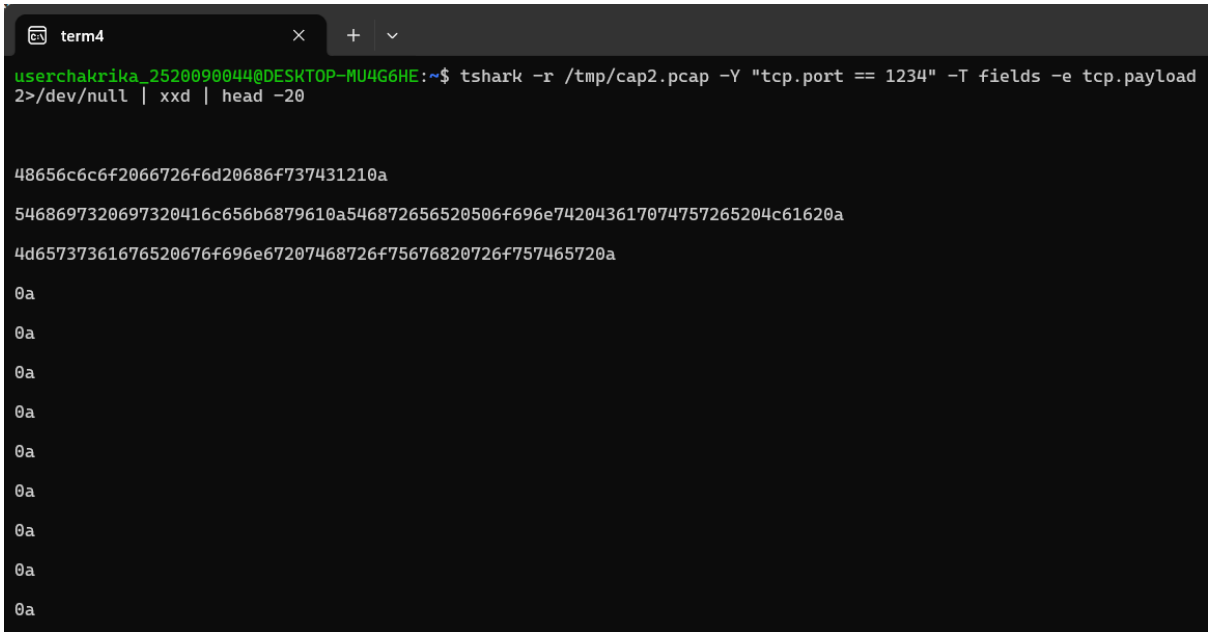
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
-rw-r--r-- 1 root root 3.5K Nov 12 09:23 /tmp/cap1.pcap
-rw-r--r-- 1 root root 3.5K Nov 12 09:23 /tmp/cap2.pcap
-rw-r--r-- 1 root root 3.5K Nov 12 09:23 /tmp/cap3.pcap
userchakrika_2520090044@DESKTOP-MU4G6HE:~$
```

Check TTL at all 3 points:

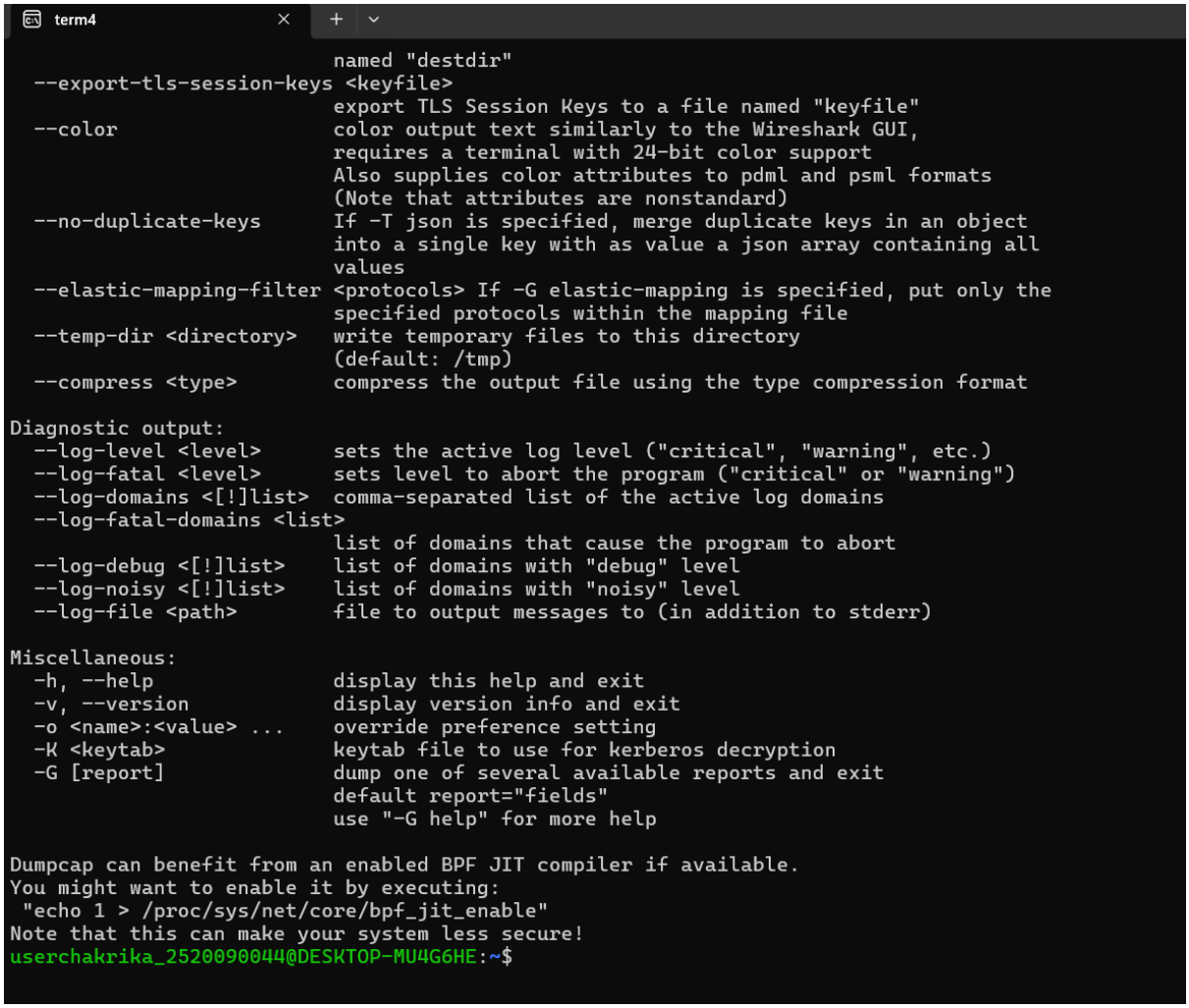
```
term4

userchakrika_2520090044@DESKTOP-MU4G6HE:~$ tshark -r /tmp/cap1.pcap -T fields -e ip.t
q -c
tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      13 63
      14 64
      13 63
      14 64
      14 63
      13 64
```

Read your actual message from router capture:



Check delay:



Check retransmissions:

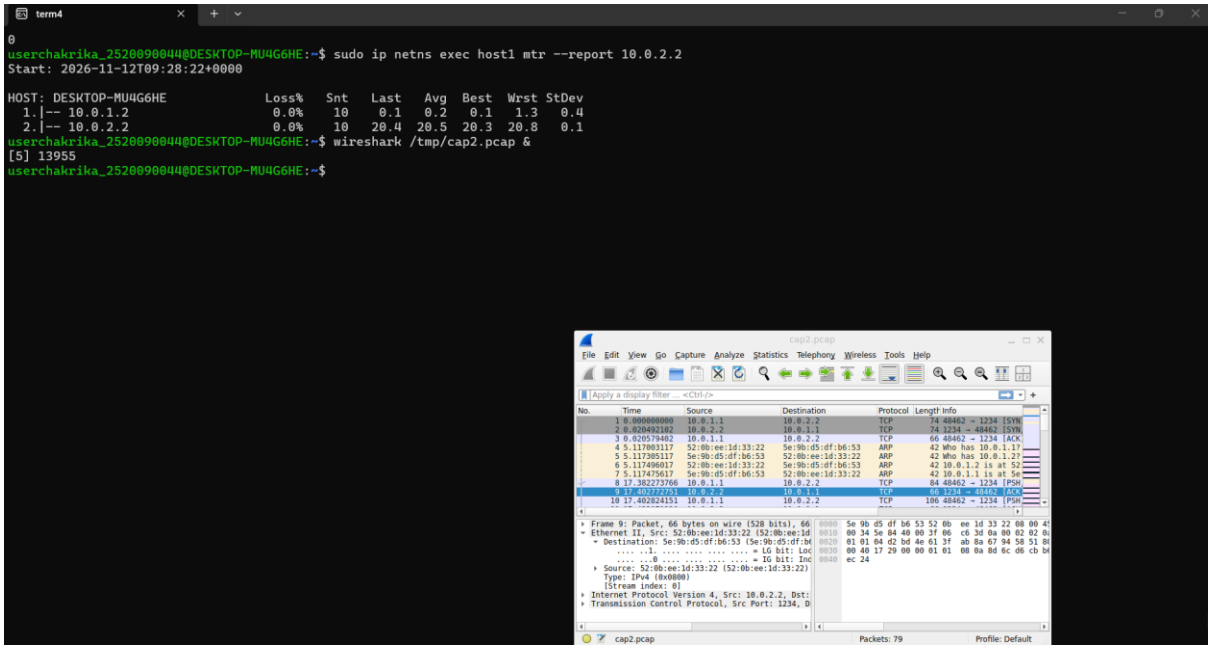
```
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ tshark -r /tmp/cap1.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
tshark -r /tmp/cap2.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
tshark -r /tmp/cap3.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
0
0
```

Check RTT:

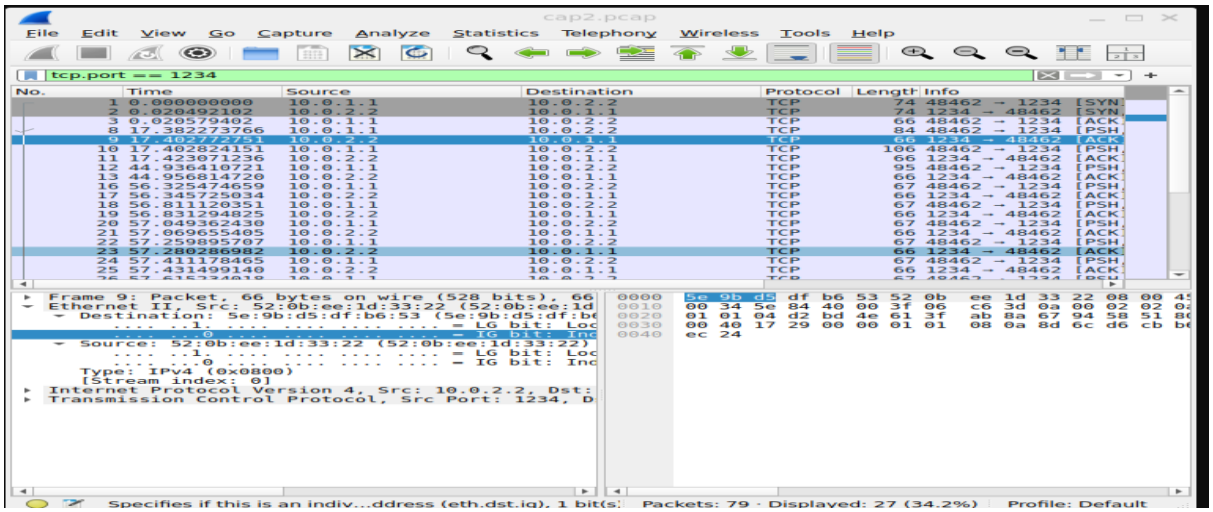
```
0
userchakrika_2520090044@DESKTOP-MU4G6HE:~$ sudo ip netns exec host1 mtr --report 10.0.2.2
Start: 2026-11-12T09:28:22+0000

HOST: DESKTOP-MU4G6HE             Loss%   Snt   Last   Avg  Best  Wrst StDev
  1.|-- 10.0.1.2                   0.0%    10    0.1   0.2   0.1   1.3   0.4
  2.|-- 10.0.2.2                   0.0%    10   20.4  20.5  20.3  20.8   0.1
userchakrika_2520090044@DESKTOP-MU4G6HE:~$
```

TERMINAL 4 — Open Wireshark



Filter your message only:



Merge all 3 captures: